

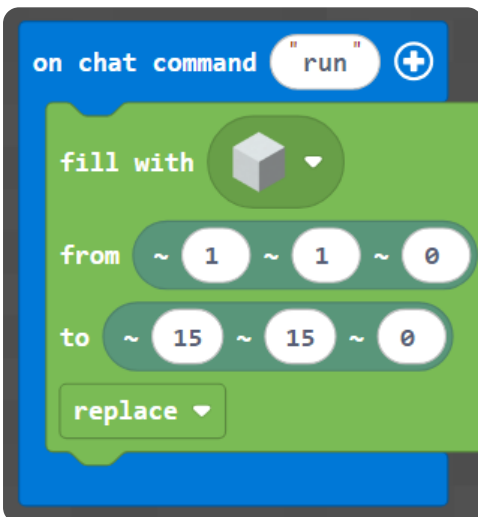
🐼 M003 Extension — English Worksheet (Advanced)

Topic: Pixel-Art Snow Golem (x, y) · **Course:** 3D Coordinates · **Level:** Advanced (Extension, after Week 3) · **Time:** about 45 minutes

A bonus picture challenge: build a whole **snow golem** — snow body, carved-pumpkin head, coal buttons, stick arms — then make it your own. Every block sits at **(x, y)** — x is how far **across**, y is how far **up**.

Build your canvas. Run this to make a blank **15 × 15** wall, then put a **red block at your feet** as your **home spot**:

🧩 Blocks



🐍 Python

```
def on_run():
    blocks.fill(WHITE_CONCRETE,
                pos(1, 1, 0),
                pos(15, 15, 0),
                FillOperation.REPLACE)
    player.on_chat("run", on_run)
```

● **Big idea:** (x, y) is counted from your **home spot**, not the world's corner. Stand on the red block *every* time you run, or the whole picture shifts.

1 · Predict 🧠

Read each set of steps. Write what you will see **and the reason**.

```
place pumpkin block at (6, 15)
place pumpkin block at (7, 15)
place pumpkin block at (8, 15)
```

Three pumpkin blocks share $y = 15$. Do they make a line going across or up? Which number stays the same, and why?


```
place black block at (8, 3)
place black block at (8, 5)
place black block at (8, 7)
place black block at (8, 9)
```

These four coal buttons share $x = 8$. Are they a row going across, or a column going up? Which number changes, and why?


```
place snow block at (5, 4)
place snow block at (6, 4)
place snow block at (5, 5)
place snow block at (6, 5)
```

These four blocks change both numbers. What shape do they make on the wall, and why?


```
place snow block at (8, 7)
place black block at (8, 7)
```

Two blocks aim for the same (x, y) . What happens on the wall — snow, black, or both? Explain the reason.

2 · Spot the Bug 🐛

Each code block below is broken. Read what it should do, fix it, then explain why the original was wrong and your fix works.

Bug A — This should make the golem's **two eyes**, one at (7, 13) and one at (9, 13). Both land in the **same spot**.

```
place black block at (7, 13)
place black block at (7, 13)
```

Hint: the two eyes are at **different** x values — one is further across than the other.

Write the fixed code:

Why was it wrong? Why does your fix work?

Bug B — A **coal button** should sit at (8, 7). The two numbers are **swapped**, so it lands in the wrong place.

```
place black block at (7, 8)
```

Hint: the **first** number is across (x), the **second** is up (y). Which one should be 8, which should be 7?

Write the fixed code:

Why was it wrong? Why does your fix work?

Bug C — A friend stood on the **wrong block** and ran correct code. The golem came out **shifted two blocks too high** — every block landed at a y two bigger than the grid says.

```
(the code matched the grid exactly)
```

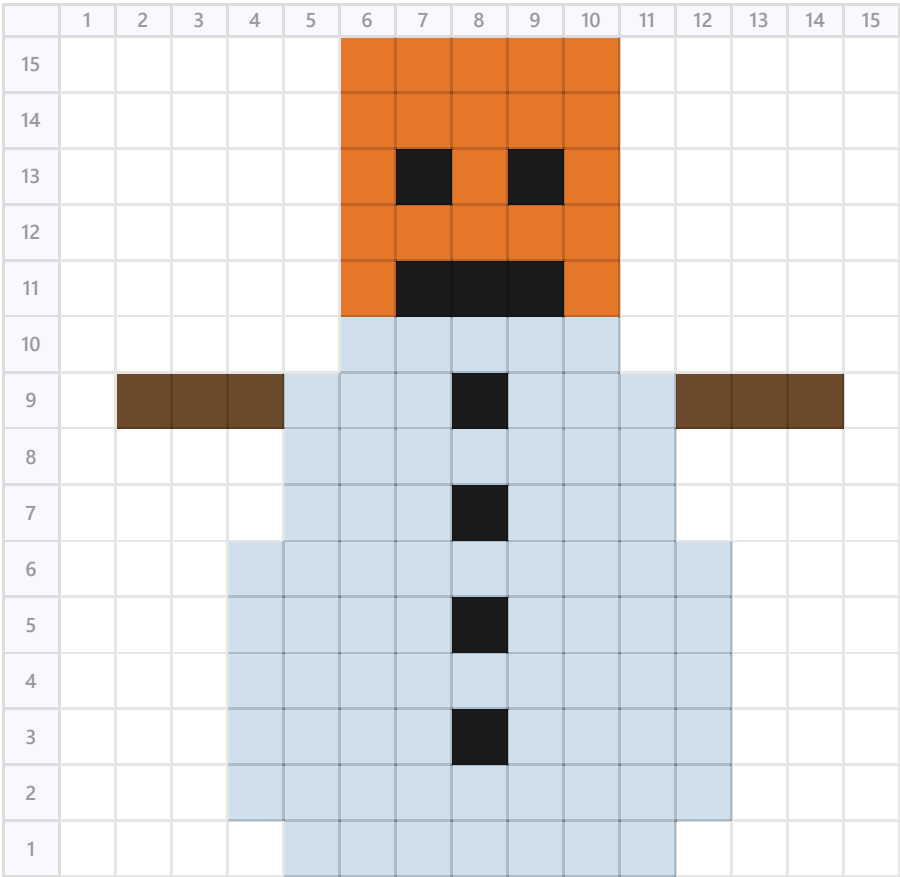
What did your friend forget to do before pressing run? Why does standing on the home spot change where every block lands?

3 · Show Your Work 📷 🎥

Stand on your **home spot** (the red block). Copy the golem block by block, reading each colored square's (x, y) — **x** across the top, **y** down the side.

Plan first. Before you place anything, find the **lowest** block and the **left-most** block of the golem and write their (x, y) here. That tells you where the picture starts.

🧊 **Snow Golem** (snow · light blue #cfe0ec, pumpkin · #e5772b, face + coal buttons · black, arms · brown #6b4a2b)



Modify challenge. After you copy the golem exactly, change **one thing** that is your own idea — swap a colour, add a scarf row, give it a longer arm, or add more buttons. Write the (x, y) of the blocks you changed, then build it.

Record **one video** — one take, no stopping (a phone is fine). Show these in order:

1 • Your code. Scroll through it. Say what each part does.

2 • Your build. Point the camera. Name the parts.

Fill the blanks:

Today I built _____.

I built it using this code: _____.

In this code I used _____.

The hardest part was _____.

That part was hard because _____.

The most fun part was _____.

Something new I learned was _____.

Write your lines here, then say them in your video.

Submit 

Send this worksheet + your video to teacher on KakaoTalk.